USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22CSE249]

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

DESIGN AND ANALYSIS OF ALGORITHMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5x2=10**
1.	**a.**	Define Algorithm?	L1	CO1	**02**
	b.	Explain about Quick sort.	L2	CO2	**02**
	c.	Discuss about minimal cost spanning tree.	L3	CO3	**02**
	d.	Define Travel Sales Person Algorithm?	L1	CO4	**02**
	e.	Distinguish P and NP problem.	L4	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain about characteristics of an algorithm?	L2	CO1	**05**
		OR			
	b.	Demonstrate about performance analysis?	L3	CO1	**05**
3.	**a.**	Explain about Linear search?	L2	CO2	**05**
		OR			
	b.	Demonstrate about Merge sort with example.	L3	CO2	**05**
4.	**a.**	Explain about single source shortest paths.	L2	CO3	**05**
		OR			
	b.	Solve a n-Queen problem using backtracking to solve placing 4 Queen problem.	L4	CO3	**05**
5.	**a.**	Explain about FIFO branch and bound problem.	L2	CO4	**05**
		OR			
	b.	Discuss about Efficiency considerations.	L3	CO4	**05**
6.	**a.**	Explain about NP complete class.	L2	CO5	**05**
		OR			
	b.	Explain about Cook's theorem.	L2	CO5	**05**
7.	**a.**	Explain about NP hard class.	L2	CO6	**05**
		OR			
	b.	Demonstrate about Satisfiability problem.	L3	CO6	**05**

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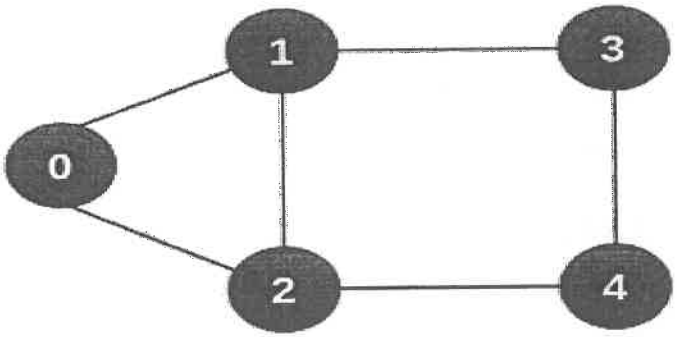
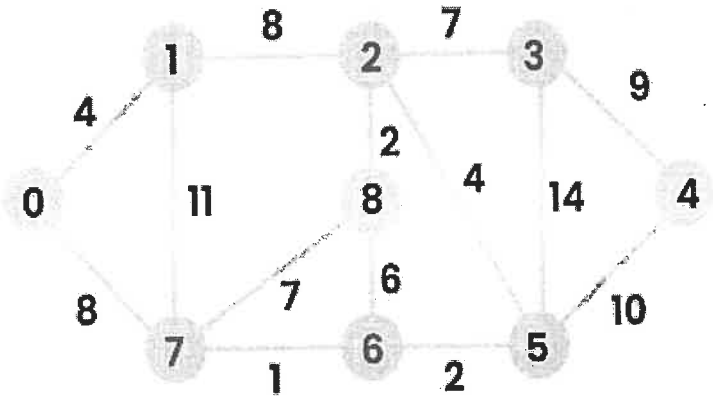
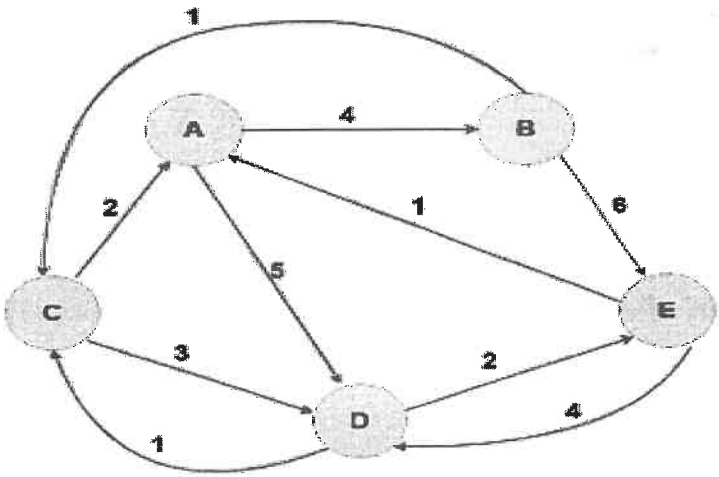
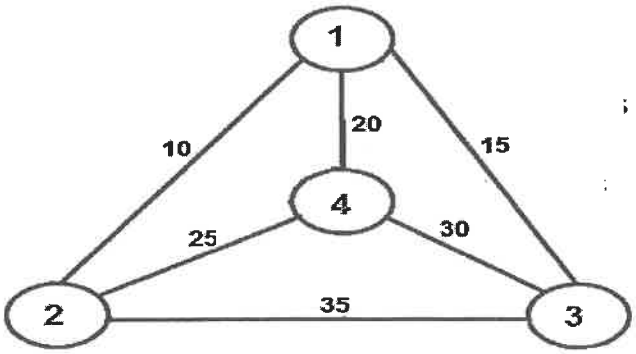
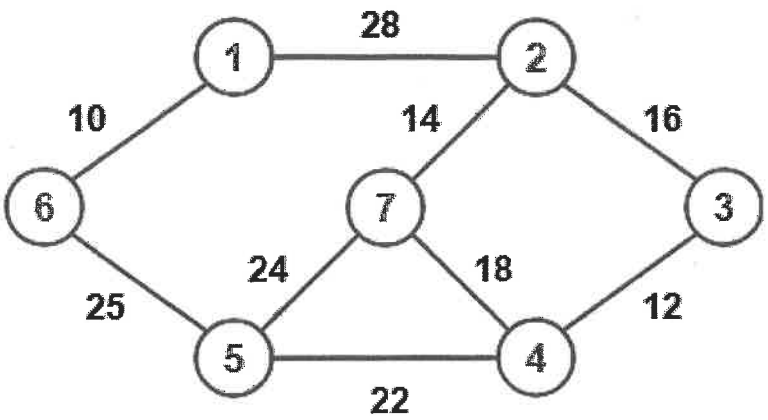
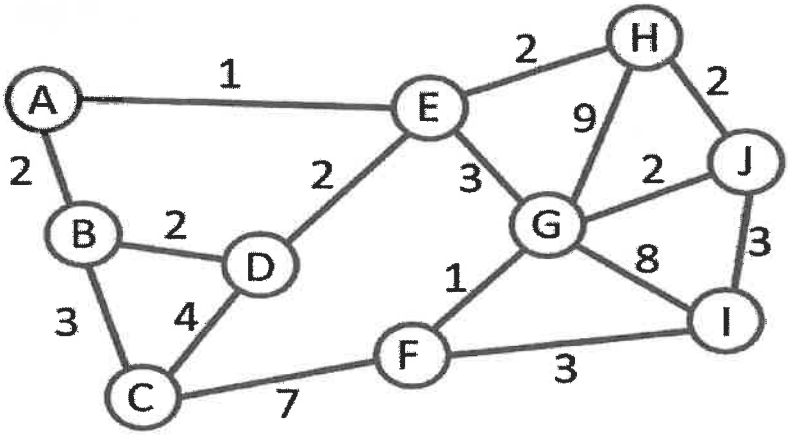
		PART-C **6 X 10=60**			
8.	**a.**	Distinguish Space complexity and Time complexity with example.	L4	CO1	**10**
		OR			
	b.	Explain in detail about Big-Oh, Omega, Theta notation.	L2	CO1	**10**
9.	**a.**	Apply quick sort algorithm in the following unsorted array. 8 7 6 1 0 9 2	L4	CO2	**10**
		OR			
	b.	Apply Breadth First Search in the following graph.  	L4	CO2	**10**
10.	**a.**	Apply Dijkstra algorithm for the following graph.  	L4	CO3	
		OR			

	b.	Apply Floyd-Warshall algorithm for the following graph. 	L4	CO3	**10**
					10
11.	**a.**	Apply 0/1 knapsack problem using branch and bound problem for the following scenario Weights: {3, 4, 6, 5} Profits: {2, 3, 1, 4} The weight of the knapsack is 8 kg	L4	CO4	**10**
		OR			
	b.	Apply Traveling salesperson problem using branch and bound problem 	L4	CO4	**10**

12.	**a.**	Apply Prims Algorithm in the following graph. 	L4	CO5	**10**
		OR			
	b.	Apply Kruskals Algorithm in the following graph. 	L4	CO5	**10**
13.	**a.**	Write a brief notes on Clique decision problem.	L3	CO6	**10**
		OR			
	b.	Write a brief note on Cook's theorem.	L3	CO6	**10**